

# Optic nerve sheath enhancement on orbital MRI in giant cell arteritis

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## ABSTRACT

**Background** Differentiating arteritic anterior ischaemic optic neuropathy (A-AION) due to giant cell arteritis (GCA) from non-arteritic anterior ischaemic optic neuropathy (NA-AION) may pose a diagnostic challenge. Our study aimed to assess the use of standard orbital MRI in distinguishing ocular manifestations of GCA from NA-AION.

**Methods** This study included 25 consecutive patients (11 GCA, 14 NA-AION) who underwent contrast-enhanced orbital MRIs within 3 months of symptom onset. Two radiologists blinded to clinical data independently evaluated MRIs for the enhancement of the optic nerve sheath (ONS) and other orbital structures.

**Results** On orbital MRI, ONS enhancement of at least one eye was more common in patients with GCA than NA-AION (64% vs 14%,  $p=0.02$ ). ONS enhancement on MRI was seen in patients with typical ophthalmologic exam findings of A-AION as well as in GCA patients with other features of ocular ischaemia (eg, retinal artery occlusion). Among patients with GCA, ONS enhancement was bilateral in six of seven cases even when visual symptoms and signs were unilateral.

**Conclusion** Patients with ocular GCA are more likely to have ONS enhancement on MRI compared with NA-AION. ONS enhancement was observed in (i) A-AION and other forms of ocular ischaemia, demonstrating the potential value of MRI in multiple orbital pathologies in GCA, and (ii) both the affected and unaffected eye, suggesting MRI may detect early subclinical ocular disease in GCA. These results highlight the potential value of adding orbital MRI to the diagnostic workup of ocular GCA.

## INTRODUCTION

In giant cell arteritis (GCA), ophthalmic manifestations ('ocular GCA') are common and can lead to the dreaded outcome of permanent vision loss.<sup>1</sup> Distinguishing GCA-related arteritic anterior ischaemic optic neuropathy (A-AION) from non-arteritic anterior ischaemic optic neuropathy (NA-AION) can sometimes be challenging especially when systemic symptoms suggestive of GCA are absent or minimal at presentation, an entity known as occult GCA.<sup>2</sup> Current diagnostic tests are helpful but have limitations; for example, temporal artery biopsy (TAB) is invasive and lacks sensitivity, and acute phase reactants are non-specific.<sup>3</sup>

MRI may be an adjunctive tool to improve the accuracy of diagnosing GCA and its ophthalmic

## WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Prior case series and cross-sectional studies using specialised, high-resolution cranial vessel wall MRI have identified changes in ocular giant cell arteritis (GCA). However, vessel wall MRI is a specialised imaging technique that is not available at many centres and is not optimised for assessing intraorbital pathology.
- ⇒ There is little data on dedicated, standard orbital imaging specifically in A-AION compared with NA-AION.

## WHAT THIS STUDY ADDS

- ⇒ Patients with A-AION and other forms of ocular ischaemia in GCA are more likely to have optic nerve sheath enhancement observed on standard orbital MRI.

## HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ This study supports the possibility that contrast-enhanced orbital MRI may be a useful adjunctive tool in the diagnostic assessment of ocular GCA.

complications. Recent case-control studies have described the enhancement of orbital structures, such as the ophthalmic artery, on vessel wall imaging in A-AION vs NA-AION when using 3-D high-resolution (eg, black-blood) vessel wall MRI of the brain, a specialised imaging technique.<sup>4–6</sup> However, vessel wall MRI is not available at many centres and is optimised for visualising contrast enhancement of the arterial walls, not intraorbital pathology.

Dedicated orbital MRI, which has modifications in acquisition plane, field of view, spatial resolution and fat suppression,<sup>7</sup> is better suited to assess the orbital structures associated with GCA, such as the optic nerve sheath (ONS) where the posterior ciliary arteries reside. In addition, the widespread availability of orbital MRI improves its clinical applicability over vessel wall imaging, which is available at limited centres. While there have been prior case series and cross-sectional studies using cranial vessel wall MRI,<sup>8,9</sup> there is little data on comprehensive, dedicated orbital imaging specifically in ocular GCA. Prior histologic studies demonstrated inflammation of the posterior ciliary arteries located in the ONS in patients with GCA, supporting the premise that ONS enhancement



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reflects GCA-related disease activity.<sup>10 11</sup> Our previous study combined vessel wall MRI with dedicated orbital imaging to compare GCA to non-GCA; in this prior study, the comparators were primarily patients with systemic manifestations without optic neuropathy or other ophthalmic diseases.<sup>6</sup> Whether dedicated orbital MRI alone (without vessel wall imaging), which is more widely available in clinical practice, can distinguish ocular GCA from another more common optic neuropathy, NA-AION, is poorly understood.

The goal of this study was to assess the use of dedicated orbital MRI in distinguishing ocular GCA, presenting with AAION and other eye manifestations, from NA-AION. We investigated the prevalence of orbital MRI changes compared with standard diagnostic measures in patients with ocular GCA vs NA-AION.

## METHODS

### Study population

This case-control study was conducted at a single institution between 2019 and 2023. Consecutive patients who met eligibility were included. Patients with contrast-enhanced orbital MRI within 3 months of visual symptom onset were included. All charts were reviewed for clinical data around the time of MRI and at least 6 months afterwards to ensure no change in clinical classification.

Ocular GCA cases were defined as patients over the age of 50 years with a diagnosis of GCA and related ophthalmic manifestations according to the neuro-ophthalmologist and rheumatologist. A diagnosis of GCA was based on TAB, clinical features of GCA and/or the 2022 ACR/EULAR classification criteria for GCA.<sup>12</sup> A positive TAB for GCA was defined as an immune cell infiltrate in the arterial wall with or without giant cells. Ophthalmologic manifestations of GCA were defined as typical visual symptoms (transient or permanent vision loss, diplopia or scotoma) and/or ophthalmologic examination suggestive of ischaemic optic neuropathy (eg, optic nerve head oedema). The appearance of the optic nerve head oedema was further characterised by whether the swelling appeared pallid, since that is often highly suspicious of GCA. The note was made of the presence of choroidal non-perfusion on intravenous fluorescein angiography (FA) and/or the presence of retinal ischaemia such as cotton wool spots and retinal arteriolar occlusions.

Controls were patients diagnosed with NA-AION and meeting the following inclusion criteria: (1) age over 50 years, (2) NA-AION diagnosed by ophthalmologists, (3) acute vision loss in one or both eyes with optic disc swelling and (4) other causes of optic disc swelling, including GCA, ruled out by appropriate testing.

This study was approved by the Institutional Review Board of the University of Pennsylvania. Written informed consent was obtained for patients who lacked a clinical indication for neuro-imaging at the time of presentation and obtained an orbital MRI for research purposes.

### Imaging acquisition

All patients underwent pre- and post-contrast enhanced orbital MRI with dedicated fat-suppressed sequences on either a 1.5-Tesla or 3-Tesla MRI system. The orbital MRI protocols included the following: fat-suppressed pre- and post-contrast axial and coronal T1-weighted turbo spin-echo sequences, fat-suppressed axial and coronal T2-weighted turbo spin-echo sequences, axial diffusion-weighted imaging of the whole brain and coronal short tau inversion recovery (STIR) imaging. 0.2 mL/kg (0.1 mmol/kg)

IV bolus of contrast-agent Dotarem (gadoterate meglumine) was infused at a rate of 2 mL/s (Princeton, NJ: Guerbet USA).

### Imaging analysis

Two neuro-radiologists (SM and RR), each with over 20 years of experience blinded to all clinical data including the final diagnosis, independently evaluated anonymised MR images. A consensus reading session was performed 6 weeks later and was used as a reference for analysis. For equivocal cases, a third blinded neuro-radiologist (RK) served as a tie-breaker.

Based on previous reports, the following four features were assessed on MRI as present or absent in each eye: (1) ONS or adjacent orbital fat enhancement,<sup>8</sup> (2) optic disc enhancement,<sup>13</sup> (3) optic nerve enhancement<sup>8</sup> and (4) optic nerve T2 hyperintensity.<sup>8</sup> For each feature, neuroradiologists scored their diagnostic confidence (either low/medium confidence or high confidence). Additional MRI sequences (eg, concurrent vessel wall imaging) were removed to minimise bias.

### Statistical analysis

To compare features between GCA and NA-AION, we used Fisher's exact or Wilcoxon's rank sum tests. Test characteristics including sensitivity, specificity and area under the ROC curve were calculated and then compared using McNemar's test. Subgroup analyses of GCA patients based on neuro-ophthalmologic examination (pallid optic disc oedema or choroidal ischaemia on FA,<sup>14</sup> erythrocyte sedimentation rate (ESR)/C-reactive protein (CRP) and glucocorticoid treatment were also performed. Cohen's kappa assessed inter-reader agreement based on the initial independent evaluations. Data was analysed using Stata 17 (College Station, TX: StataCorp LLC).

## RESULTS

### Description of the study population

In total, 25 patients were included: 11 ocular GCA and 14 NA-AION (table 1). GCA patients were older and more likely to have systemic symptoms and elevated ESR/CRP compared with NA-AION (table 1). Among GCA patients, all 11 underwent TAB, nine of whom were positive for GCA. Among NA-AION patients, one patient underwent TAB and was negative for GCA. The 2022 ACR/EULAR criteria for GCA were met in 11 of 11 GCA patients, including two patients with negative TAB.

Ophthalmologic examination revealed that patients with GCA had similar visual acuity in the worse eye (median logMAR 0.1 in GCA vs 0.1 in NA-AION,  $p=0.24$ ) and, as expected, pallid disc oedema was more prevalent in the GCA group, whereas hyperemic disc oedema was common in NA-AION (table 1). Among the 11 ocular GCA patients, nine patients had exam findings consistent with A-AION, one patient had bilateral transient vision loss and normal ophthalmic exam and one patient had right monocular transient vision loss but the exam was not performed at presentation (for the latter patient, ophthalmic examination 6 months later was normal). The typical exam signs of GCA (defined as pallid disc oedema or choroidal ischaemia on FA) were present in six of the nine A-AION patients vs none of the NA-AION patients ( $p<0.01$ ). The remaining three A-AION patients had vision loss with abnormal fundus exams significant for either non-pallid disc oedema with branch retinal artery occlusion ( $n=1$ ) or optic disc atrophy ( $n=2$ ) (online supplemental figure 1).

**Table 1** Baseline patient characteristics by diagnosis

|                                                      | GCA<br>(n=11)  | NA-AION<br>(n=14) | P value |
|------------------------------------------------------|----------------|-------------------|---------|
| <b>Demographics</b>                                  |                |                   |         |
| Age (year)                                           | 75 (69–82)     | 62 (56–67)        | <0.01   |
| Female sex                                           | 7 (64%)        | 4 (29%)           | 0.12    |
| <b>Clinical presentation</b>                         |                |                   |         |
| Systemic symptoms present                            | 11 (100%)      | 2 (14%)           | <0.01   |
| Unilateral visual symptoms                           | 6 (55%)        | 14 (100%)         | <0.01   |
| Bilateral visual symptoms                            | 5 (45%)        | 0 (0%)            |         |
| Transient visual obscurations present                | 9 (82%)        | 0 (0%)            | <0.01   |
| Met 2022 ACR/EULAR classification criteria ≥6 points | 11 (100%)      | n/a               | n/a     |
| <b>Ophthalmologic exam*</b>                          |                |                   |         |
| Visual acuity (logMAR)                               | 0.1<br>(0–0.4) | 0.1<br>(0.1–1.0)  | 0.24    |
| Relative afferent pupillary defect                   | 4 (36%)        | 10 (71%)          | 0.12    |
| Extra-ocular motility restriction                    | 0 (0%)         | 1 (7.1%)          | 1.00    |
| Pallid disc oedema                                   | 5 (45%)        | 0 (0%)            | 0.01    |
| Hyperemic disc oedema                                | 2 (18%)        | 11 (79%)          | <0.01   |
| Cotton wool spots                                    | 1 (9.1%)       | 1 (7.1%)          | 1.00    |
| Retinal artery occlusion                             | 1 (9.1%)       | 0 (0%)            | 0.46    |
| Choroidal ischaemia on fluorescein angiography       | 1 (9.1%)       | 0 (0%)            | 0.46    |
| <b>Laboratory data</b>                               |                |                   |         |
| ESR elevated†                                        | 9 (82%)        | 1 (7.1%)          | <0.01   |
| ESR value (mm/hr)                                    | 80 (53–109)    | 9 (5–11)          | <0.01   |
| CRP elevated‡                                        | 11 (100%)      | 0 (0%)            | <0.01   |
| CRP value (mg/dL)                                    | 5.2 (2.2–16)   | 0.5 (0.4–0.6)     | <0.01   |
| Positive temporal artery biopsy                      | 9 (82%)        | 0 (0%)            | <0.01   |
| <b>Treatment prior to MRI</b>                        |                |                   |         |
| Delay between symptom onset and MRI (days)           | 7 (1–28)       | 28 (10–44)        | 0.05    |
| Received steroids prior to MRI                       | 10 (91%)       | 0 (0%)            | <0.01   |
| Pre-MRI prednisone duration (days)                   | 5 (1–60)       | 0 (0–0)           | <0.01   |
| Pre-MRI prednisone dose (mg)                         | 60 (40–1000)   | 0 (0–0)           | <0.01   |

All values are expressed as n (%) or median (IQR). Systemic symptoms were defined as the presence of one or more of the following: headache, jaw pain, scalp tenderness, temporal artery tenderness or constitutional symptoms. Visual acuity represents the best-corrected visual acuity of the worst eye for each patient, converted to logMAR. Temporal artery biopsy was considered positive when containing immune cell infiltrate in the intima, media and/or adventitia with or without giant cells.

\*Presence of exam findings in at least one eye.

†Elevated ESR defined as ≥50 mm/hour.

‡Elevated CRP defined as ≥1 mg/dL.

ACR/EULAR, American College of Rheumatology/European Alliance of Associations for Rheumatology; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; GCA, giant cell arteritis; logMAR, Logarithm of the Minimum Angle of Resolution; NA-AION, non-arteritic anterior ischaemic optic neuropathy.

### Orbital MRI enhancement patterns between GCA and NA-AION

Of the four orbital structures evaluated on MRI, only ONS enhancement was significantly different between groups (figure 1). There was no difference in the presence of optic disc or optic nerve enhancement or optic nerve T2 hyperintensity on MRI between groups. We subsequently focused analyses on ONS enhancement.

ONS enhancement was observed in six of nine patients with A-AION, one of two patients with other ocular GCAs and two of 14 patients with NA-AION. When ONS enhancement was

observed in GCA, it was almost always bilateral. Among 20 eyes from 10 GCA patients with available ophthalmologic exam data, ONS enhancement was often observed even in clinically unaffected eyes, including two patients with monocular exam findings who had bilateral ONS enhancement (figure 2). Example images of ONS enhancement on GCA vs NA-AION are shown in figure 3.

### Diagnostic performance of optic nerve sheath enhancement for GCA

For the diagnosis of A-AION, ONS enhancement on MRI had a sensitivity of 67%, specificity of 86% and ROC area of 0.76 (table 2). When comparing only TAB-positive GCA patients with NA-AION patients, the diagnostic performance of ONS enhancement improved (table 2). For the diagnosis of A-AION, the performance of GCA-specific ophthalmologic exam findings (pallid disc oedema on fundus exam or choroidal hypoperfusion on FA) vs ONS enhancement on MRI was not significantly different (for A-AION, the sensitivity of exam 67% vs MRI 67%,  $p=0.99$ ; specificity of exam 100% vs MRI 86%,  $p=0.16$ ).

There was a moderate inter-reader agreement for ONS enhancement ( $\kappa=0.43$ , 72% agreement,  $p=0.01$ ). Diagnostic confidence was high for all structures, including ONS enhancement (online supplemental figure 2).

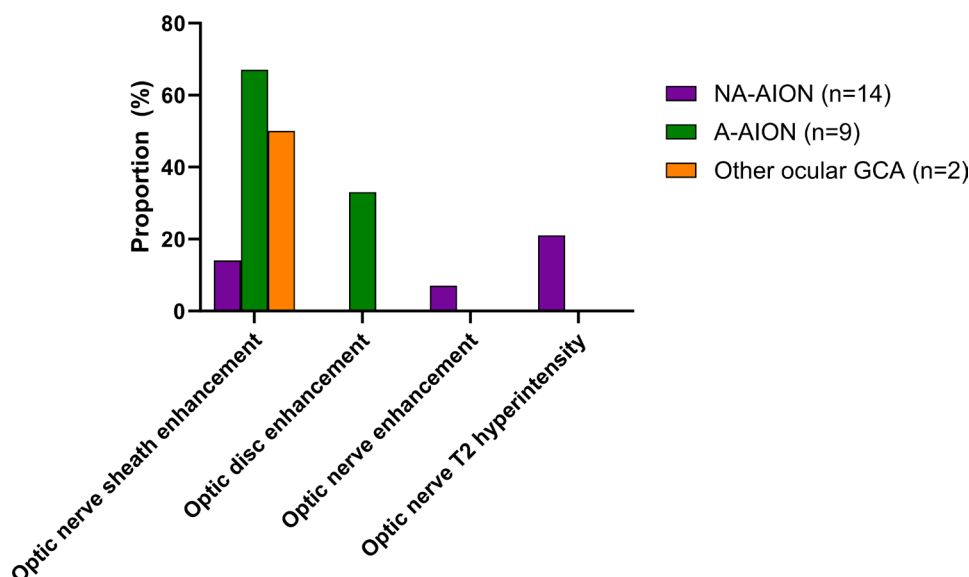
### MRI findings within subgroups of GCA patients

Among the six patients with GCA-specific ophthalmologic exam findings (pallid disc oedema or choroidal ischaemia on FA), ONS enhancement was present in four (67%) patients. Among the five GCA patients who lacked these typical ophthalmologic findings, ONS enhancement was present in three (60%) patients. Of these three patients, one patient had signs of retinal ischaemia with a positive TAB, one patient had optic disc pallor and positive TAB and one patient had a normal ophthalmologic exam, but transient vision loss in both eyes and a positive TAB.

Within the GCA group, 10 of 11 (91%) patients received prednisone prior to undergoing orbital MRI. There was no difference in the presence of ONS enhancement based on the duration or dose of prednisone. The presence of ONS enhancement was also not associated with days since the onset of visual symptoms ( $\leq 7$  days vs  $> 7$  days), peak ESR or peak CRP values.

### DISCUSSION

Ophthalmologic manifestations of GCA are an emergency, and prompt diagnosis and treatment are critical to prevent irreversible visual loss. Many patients with ocular GCA present to the emergency department where immediate evaluation by an ophthalmologist is not always available. This study suggests that dedicated orbital MRI contains useful information in identifying GCA when combined with current diagnostic approaches. While eye examination has tremendous value in distinguishing various causes of visual loss, adding orbital MRI to assess for ONS enhancement may contain additive diagnostic information and often may be easily attained since patients with sudden visual loss may already undergo MRI scans to rule out other aetiologies (eg, emboli) or may be evaluated in centres without rapid access to an ophthalmologist. Specifically, our results demonstrate that patients with ocular GCA are more likely to have ONS enhancement on MRI compared with patients with NA-AION. Other MRI findings previously reported, such as optic disc and optic nerve enhancement, were infrequently seen and not significantly different between the two groups. Orbital MRI also detected abnormalities in the contralateral, unaffected eye in patients

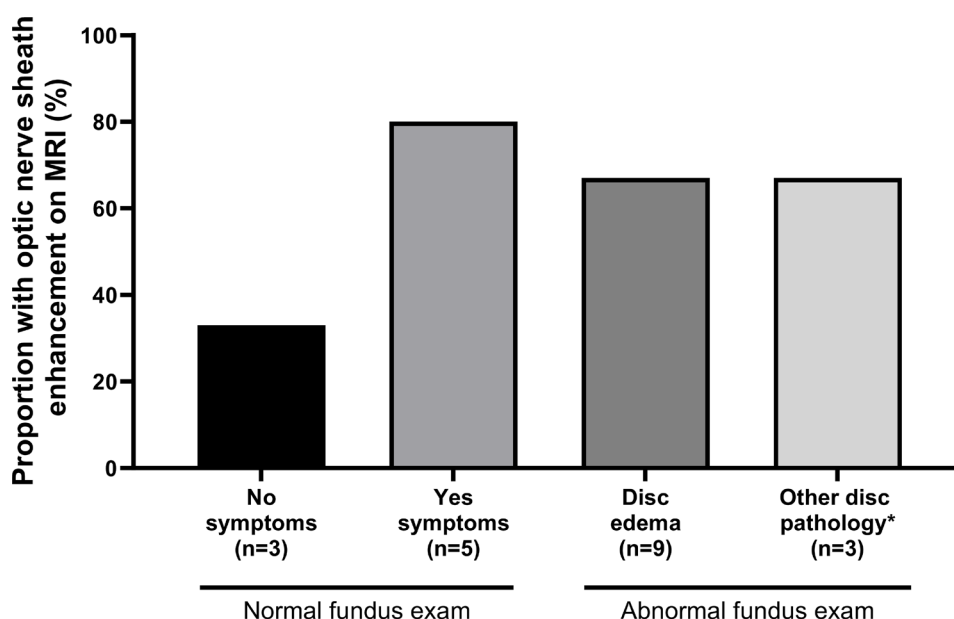


**Figure 1** Prevalence of orbital MRI findings in patients with giant cell arteritis vs non-arteritic anterior ischaemic optic neuropathy. The proportion with each MRI finding by diagnosis. Among the four MRI characteristics scored, only optic nerve sheath enhancement was significantly greater in GCA vs NA-AION (64% vs 14%,  $p=0.02$ ). A-AION, arteritic anterior ischaemic optic neuropathy; GCA, giant cell arteritis; MRI, magnetic resonance imaging; NA-AION, non-arteritic anterior ischaemic optic neuropathy.

with GCA, suggesting MRI may detect early subclinical ocular disease since the progression to binocular vision loss is common in GCA if treatment is delayed. Together, these findings support the notion that orbital MRI may have additive value in establishing the diagnosis of GCA.

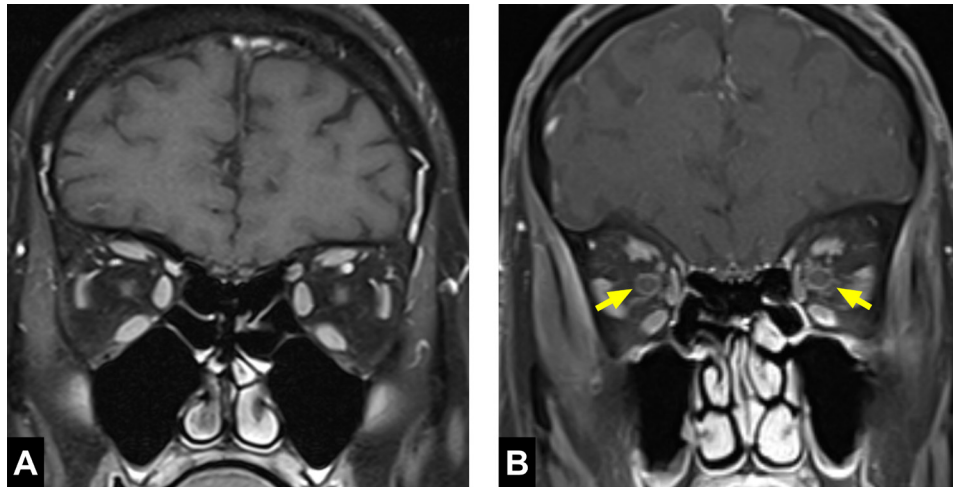
Previous studies of MRI in ocular GCA relied on high-resolution vessel wall MRI, which may have limited clinical translation at present due to its lack of availability at many centres.<sup>4 5 15</sup> Although one previous study focused on the ophthalmic artery, ONS enhancement was a secondary imaging endpoint and was able to identify A-AION with high accuracy.<sup>5</sup>

Imaging the ophthalmic artery wall is technically challenging due to its small calibre and tortuous course.<sup>16 17</sup> In contrast, our study investigated standard orbital MR sequences and ONS enhancement, which is technically easier to image and assess compared with the ophthalmic artery wall. The widespread availability of the standard orbital MRI improves its clinical applicability over vessel wall imaging. In contrast to previous studies evaluating combined orbital with vessel wall MRI in GCA vs non-GCA,<sup>6 9</sup> this study examined orbital MRI alone (after removing the vessel wall sequences), focused on the subset with ocular GCA including a more comprehensive description and analysis



**Figure 2** Optic nerve sheath enhancement on MRI by ophthalmologic symptoms and exam findings in affected and unaffected eyes of giant cell arteritis. Among 10 patients with ocular giant cell arteritis who had available funduscopic exam data, optic nerve sheath enhancement was observed even in the contralateral, unaffected eye which had a normal exam or was asymptomatic. In a majority of cases, MRI enhancement was bilateral even though visual symptoms and signs were unilateral, supporting the possibility that MRI may detect early, subclinical ocular disease in giant cell arteritis. \*Other disc pathologies included branch retinal artery occlusion, optic disc pallor and optic disc atrophy.





**Figure 3** Examples of orbital MR images. All images are coronal post-contrast T1-weighted fat-saturated orbital MR images. (A) Normal orbital MRI in a patient with non-arteritic anterior ischaemic optic neuropathy of the left eye. Optic nerve sheath enhancement was absent from both eyes. (B) Orbital MRI revealed optic nerve sheath enhancement (yellow arrows) in both eyes of a giant cell arteritis patient with arteritic anterior ischaemic optic neuropathy. This patient had bilateral pallid disc oedema on a fundus exam and positive temporal artery biopsy.

of ophthalmic manifestations and used NA-AION (as opposed to patients with headaches or other systemic manifestations) for the comparator group. Comparison to NA-AION addresses a specific question relevant to clinical practice since NA-AION is a more prevalent ischaemic optic neuropathy than A-AION and can sometimes be difficult to distinguish on ophthalmologic exams. A previous study involving 13 patients combined conventional contrast-enhanced brain and orbital MRI to evaluate seven TAB-positive vs six TAB-negative patients and found that evaluating for both optic nerve sheath and intracranial internal carotid artery (ICA) wall enhancement improved the diagnostic accuracy<sup>9</sup>; however, subsequent studies, which included larger sample sizes ( $n=105$  and  $n=168$ ) and used dedicated vascular imaging, found a much lower prevalence of intracranial ICA involvement (4–10%),<sup>16 18</sup> suggesting the addition of dedicated imaging of the intracranial ICA likely has low diagnostic value and was therefore not included in our study. Lastly, unlike previous studies,<sup>13</sup> this study found a low prevalence of optic disc enhancement in GCA, potentially due to differences in the imaging technique or study population.

Previous post-mortem and histopathological reports support the possibility that MRI enhancement of the ONS reflects inflammatory changes in GCA. In a post-mortem study of 12 patients with GCA, 75% of posterior ciliary and ophthalmic arteries had pathologic changes consistent with vasculitis, and

the most severe findings were observed in the posterior ciliary arteries located around the ONS,<sup>10</sup> the site of MRI enhancement observed in our study. One case report described a patient with A-AION secondary to GCA (diagnosed by positive TAB and bilateral pallid disc oedema) who was found to have bilateral ONS enhancement on MRI; biopsy of the ONS revealed pan-mural arterial inflammation with giant cells, providing pathologic evidence that ONS enhancement on MRI visualises GCA-related inflammation.<sup>11</sup> It is also important to note that ONS enhancement can be seen in other conditions, such as malignancies, infections and other inflammatory diseases.<sup>19</sup> As with all diagnostic tests, thoughtful interpretation and appropriate clinical contextualisation of MRI will be imperative.

Our findings suggest that orbital MRI may be useful in facilitating early and accurate diagnosis, especially when combined with ophthalmic examination. The classic exam finding for GCA that distinguishes it from NA-AION is pallid optic disc oedema on funduscopy or choroidal ischaemia on FA, but this is reported in around 50% of GCA patients,<sup>20</sup> similar to our study. When visual symptoms or funduscopy examination findings are equivocal—such as the presence of non-pallid disc oedema or vague complaints of blurry vision or transient visual loss without evidence of optic nerve or retinal ischaemia or when there is a lack of systemic signs and symptoms, then orbital MRI may serve as a useful adjunct to the diagnostic work-up of GCA.

Our study has a few limitations. First, 92% of the GCA patients in our cohort were treated with prednisone prior to undergoing MRI compared with none of the NA-AION patients. However, the expected effect of glucocorticoid therapy would be to mask any inflammatory enhancement on MRI and therefore would bias to the null. Second, NA-AION patients experienced a greater delay between the symptom onset and MRI compared with GCA patients; however, even among NA-AION patients who underwent MRI within a few days of symptom onset, no MRI enhancement was observed. Third, the sample size is small (25 patients) and is derived from only a single centre; prior to clinical implementation, additional studies are needed to externally validate these results. Additionally, by design, the GCA cases and NA-AION controls that were enrolled in the study had a clear and definitive diagnosis. Patients with equivocal presentations

**Table 2** Diagnostic performance of optic nerve sheath enhancement on MRI in at least one eye for arteritic anterior ischaemic optic neuropathy

|                                                                                                                                                                                                                         | All A-AION (n=9) vs NA-AION (n=14) | TAB-confirmed A-AION (n=8) vs NA-AION (n=14) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------|
| Accuracy, median (IQR)                                                                                                                                                                                                  | 78% (56%–93%)                      | 82% (60%–95%)                                |
| Sensitivity, median (IQR)                                                                                                                                                                                               | 67% (30%–93%)                      | 75% (35%–97%)                                |
| Specificity, median (IQR)                                                                                                                                                                                               | 86% (57%–98%)                      | 86% (57%–98%)                                |
| ROC area, median (IQR)                                                                                                                                                                                                  | 0.76 (0.56–0.93)                   | 0.80 (0.60–0.95)                             |
| A-AION, arteritic anterior ischaemic optic neuropathy; MRI, magnetic resonance imaging; NA-AION, non-arteritic anterior ischaemic optic neuropathy; ROC, receiver operating characteristic; TAB, temporal artery biopsy |                                    |                                              |

of ocular GCA, which pose a greater diagnostic challenge, were not represented in our study population. In addition, the subset of NA-AION controls that were eligible for this study were those that warranted neuro-imaging and may reflect a specific subgroup of NA-AION. Lastly, diffusion restriction of the optic nerve was not included since the diffusion-weighted imaging sequences were not always optimised to assess the optic nerve in this cohort. Given that there have been conflicting data on the presence of diffusion restriction of the optic nerve in GCA,<sup>8 21</sup> future studies using optimised DWI sequences are needed.

In summary, GCA patients are more likely to have ONS enhancement on orbital MRI compared with NA-AION patients, suggesting that dedicated orbital MRI can have a useful adjunctive role in distinguishing between GCA and NA-AION. Although larger studies are needed to validate these results, our data supports the possibility that MRI may transform the clinical management of this vision-threatening condition.

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**Patient consent for publication** Not applicable.

**Ethics approval** This study involved human participants and was approved by the Institutional Review Board of the University of Pennsylvania (#843171). Informed consent was obtained for patients who lacked a clinical indication for neuro-imaging at the time of presentation and obtained an orbital MRI for research purposes.

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